

Estimating weather impact on the duration of construction activities

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Abstract: Weather conditions can have an adverse impact on the duration and cost of construction activities. Quantifying this impact is, clearly, valuable to contractors for preparing realistic schedules, cost estimates, and reliable bids. Productivity loss due to the impact of weather on construction activities can be either partial or complete; partial loss is generally attributed to reduced labor productivity and complete to work stoppage which interrupts those activities. This paper presents an automated decision support system, named WEATHER, for estimating the combined effect of reduced labour productivity and work stoppage caused by adverse weather conditions on construction sites. The system provides estimates of construction productivity, activity durations, and weather patterns that facilitate the application of risk analysis in planning and scheduling. The system is portable and can be used in all cities across Canada where weather data are available. WEATHER is flexible and can be used in a default mode or in a user-input mode to estimate the duration of construction activities. In the default mode, the system provides default threshold values for key weather parameters. In the user-input mode, the system requires the user to define these thresholds. The system can be used to estimate weather impact before and during construction, for scheduling, and after construction for claim analysis. A numerical example is analyzed to illustrate the use of the system and demonstrate its capabilities.

Key words: weather impact, construction productivity, planning and scheduling, decision support system.

Résumé : Les conditions météorologiques peuvent avoir un impact défavorable sur la durée et le coût des activités de construction. La quantification de cet impact est importante pour les entrepreneurs afin de préparer des plans réalistes, des estimations de coût, et des offres fiables. La perte de productivité due à l'impact météorologique sur les activités de construction peut être partielle ou complète; la perte partielle est généralement attribuée à une réduction de productivité de la main-d'oeuvre, et celle complète à l'arrêt de travail qui suspend ces activités. Cet article présente un système automatisé de support de décision, nommé WEATHER, pour estimer l'effet combiné de réduction de productivité de la main-d'oeuvre et l'arrêt du travail causé par des conditions météorologiques défavorables sur les sites de construction. Le système offre des estimations de la productivité de construction, des durées d'activité et des tendances de la météo, ce qui facilite l'application de l'analyse de risque dans la planification et programmation. Le système est portable et peut être utilisé dans toutes les villes canadiennes où des données météorologiques sont disponibles. Le système WEATHER est flexible et peut être utilisé selon un mode par défaut ou selon un mode d'entrée par l'utilisateur. Dans le mode par défaut, le système offre des valeurs seuils pour des paramètres météorologiques clés. Dans le mode d'entrée par l'utilisateur, le système nécessite la définition de ces seuils par l'opérateur. Le système peut être utilisé pour estimer l'effet de l'impact de la météo avant et durant la construction, pour la planification, et après la construction pour l'analyse des réclamations. Un exemple numérique est analysé pour illustrer l'utilisation du système et démontrer ces capacités.

Mots clés : impact météorologique, productivité de construction, planification et programmation, système de support de décision.

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Introduction

Construction projects, in general, are executed in an outdoor environment, and therefore are affected by weather conditions. Weather impact was reported to be one of the main factors causing delay and cost overruns on construction projects

(Baldwin et al. 1971; Koehn and Meilhede 1981; Laufer and Cohenca 1990). Considerable number of construction activities are sensitive to weather conditions. Benjamin et al. (1973) suggested that almost 50% of construction activities are sensitive to weather conditions. The impact of weather on construction activities can be in the form of reduced labour productivity and (or) work stoppage. Reduced labour productivity is generally attributed to reduced human performance due to heat or cold stresses resulting from the combined effect of temperature, humidity, and wind velocity. Weather-related work stoppage is attributed either to the inability of construction personnel to work under severe weather conditions of heavy rain, snow, and (or) gusting winds or simply to compliance with safety regulations in such adverse weather conditions.

A number of studies have been conducted to establish the relation between labour productivity and weather conditions

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for electrical work (National Electrical Contractors Association 1974), masonry construction (Grimm and Wagner 1974; Sanders and Thomas 1991), equipment and manual tasks (U.S. Army Cold Regions Research and Engineering Laboratory 1986), and general construction (Koehn and Brown 1985). The latter provides an industry-wide average considering a number of trades (manual excavation, equipment excavation, steel erection, masonry, electrical, and carpentry). Others (Cantwell 1987; Smith and Hancher 1989) considered the impact of weather on work stoppage and established daily rainfall thresholds that would cause the stoppage and interruption of construction activities.

A system that provides a comprehensive estimate of weather impact (i.e., accounting for reduced labour productivity and interruptions causing work stoppage) on construction productivity is of practical value. It assists in quantifying weather impact on productivity and duration of construction activities, which facilitates the preparation of realistic schedules, cost estimates, and reliable bids. Benjamin et al. (1973) proposed a simulation model that integrates the interruptive effect of weather in scheduling. The model simulates construction duration by making daily work/no-work decisions according to the historical hourly weather data and the sensitivity of activities to temperature, precipitation, and wind. Moselhi and Nicholas (1990) proposed a hybrid expert system for construction planning and scheduling that considers the impact of reduced labour productivity due to weather. Based on adjusted productivity factors, the system generates automatically a revised or "as possible" schedule. The latter work has been expanded in the development of a stand-alone decision support system, WEATHER, for estimating weather impact on construction productivity and durations of activities (Moselhi et al. 1995).

This paper presents recent developments aimed at expanding the applications of WEATHER (Moselhi et al. 1995) to include (i) weather impact in the form of interrupted working hours or days due to precipitation and (or) gusting winds and (ii) estimation of a joint productivity factor that accounts for the combined impact of weather on labour productivity and work interruption. The system is currently limited to considering impact of weather on electrical work, masonry construction, outdoor manual and equipment tasks, and general construction, as described earlier. The paper also provides a detailed description of WEATHER.

WEATHER

WEATHER is an automated decision support system developed for estimating the impact of weather on construction productivity. Weather impact is quantified based on either an industry-wide average performance on general construction work or industry-specific performance on specialized work, which includes electrical work, masonry work, general manual tasks, and equipment tasks. WEATHER is coded using Visual C++ for Windows and provides an application that runs on Microsoft Windows. The system can be used as a stand-alone module for estimating the impact of weather on productivity and on duration of construction activities, making it useful in developing realistic schedules and in preparing, analyzing, and negotiating claims that involve adverse weather conditions. The system can also be used as a submodule in an integrated

scheduling system that considers the additional impact of other factors on productivity, such as overtime, learning curve, site congestion, and change orders. WEATHER has three main modules: Consultant, Estimator, and Analyzer, supported by a user interface and a database as shown in Fig. 1.

Consultant

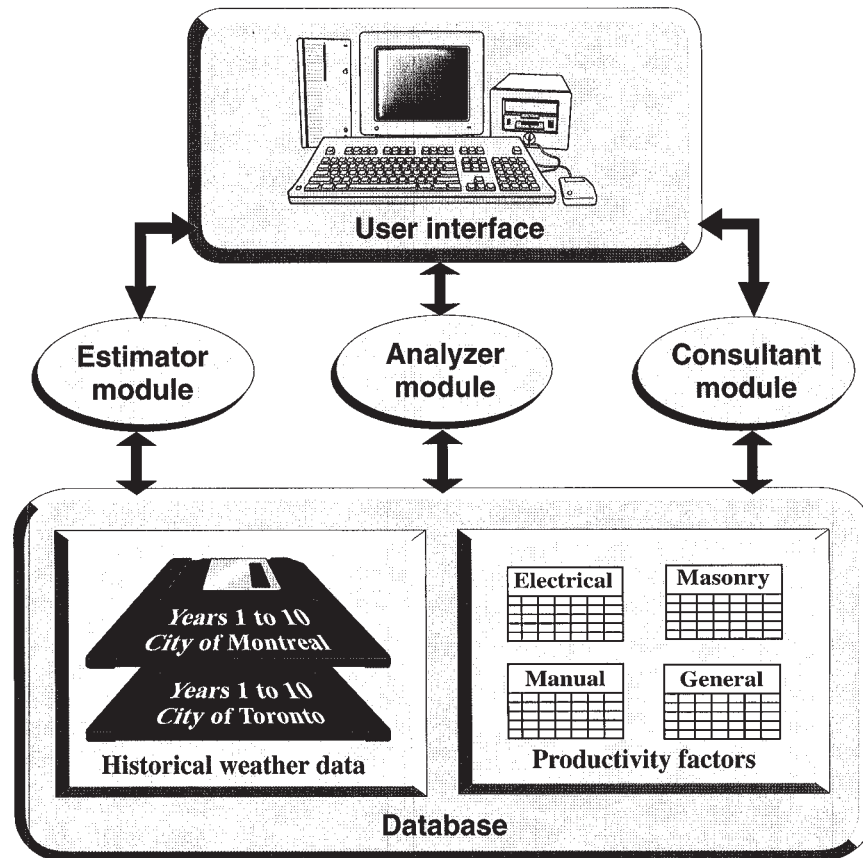
Consultant provides the user with a graphical illustration of the impact of key weather parameters (e.g., temperature, humidity, and wind) on labour productivity for different categories of construction tasks or activities. In order to study the impact of temperature and wind velocity on productivity, for example, the user has to identify the type of construction activity and the humidity level. The system then presents a graph that illustrates the change in productivity as temperature and wind velocity vary (see Fig. 2). In Consultant, a set of productivity factors for the activity being considered is calculated based on the models described by Grimm and Wagner (1974), National Electrical Contractors Association (1974), U.S. Army Cold Regions Research and Engineering Laboratory (1986), and Koehn and Brown (1985).

Estimator

Estimator is designed for estimating modified activity duration, accounting for the impact of weather and considering both reduced labour productivity and hours or days of work stoppage. In considering weather-related work stoppage, Estimator operates in two modes: a default mode and a user-input mode. In the default mode, the system uses default threshold values for precipitation and wind speed in order to estimate activity durations. In the user-input mode, the user can simply override the system and specify threshold values as shown in Fig. 3.

In the user-input mode, the user is required to input pertinent data in two dialogue windows: Site Parameters and Activity Parameters. As shown in Fig. 3, the Site Parameters dialogue window requires the user to input the location of construction site and threshold values of rainfall, snowfall, and wind velocity, beyond which the activity work should be interrupted. As shown in Fig. 4, the Activity Parameters dialogue window requires the user to identify the type of construction work or task, scheduled start date, and estimated activity duration without weather impact. In addition to these two dialogue windows, WEATHER allows the user to input and save actual recorded weather data on site by using the User Input Weather Data dialogue window shown in Fig. 5. These actual weather data enable the user to analyze the effect of weather conditions on construction duration and, accordingly, provide support in preparing construction claims.

In the Site Parameters dialogue window, WEATHER also allows the user to select one of two possible types of work interruption (i.e., A and B) as shown in Fig. 3. Type A is when the work being considered can resume immediately after an interruption period. In this type, interrupted work hours are considered equal only to those when adverse weather conditions were found to exceed the user-specified threshold values of precipitation and (or) wind. Type B can be selected if interruption hours exceed the direct loss calculated in type A. In this type, the user can specify if the interruption should be a half day or a whole day according to the time of interruption (i.e., morning or afternoon) and duration of interruption

Fig. 1. WEATHER components.

(i.e., number of hours) as shown in Fig. 3. This flexibility in modelling work interruptions is practical and useful. For some activities such as excavation, the lost working hours due to rain may extend beyond the raining hours because of the difficulty associated with operation of earthmoving equipment, particularly cohesive soils and common earth. The impact of weather on such activities can best be modelled by selecting type B and specifying additional interruption period as shown in Fig. 3.

Analyzer

Analyzer provides a statistical analysis of weather data, including wind velocity, temperature, and precipitation within a specified execution period of an activity. The analysis is based on 10 years of historical weather data pertaining to the city being considered. Figure 6, for example, shows the statistical analysis of raining hours within the specified execution period of the activity. Analyzer helps the user to (i) recognize likely weather conditions that may prevail over the duration of scheduled construction operations and (ii) analyze the estimates made by Estimator for productivity factors and activity durations.

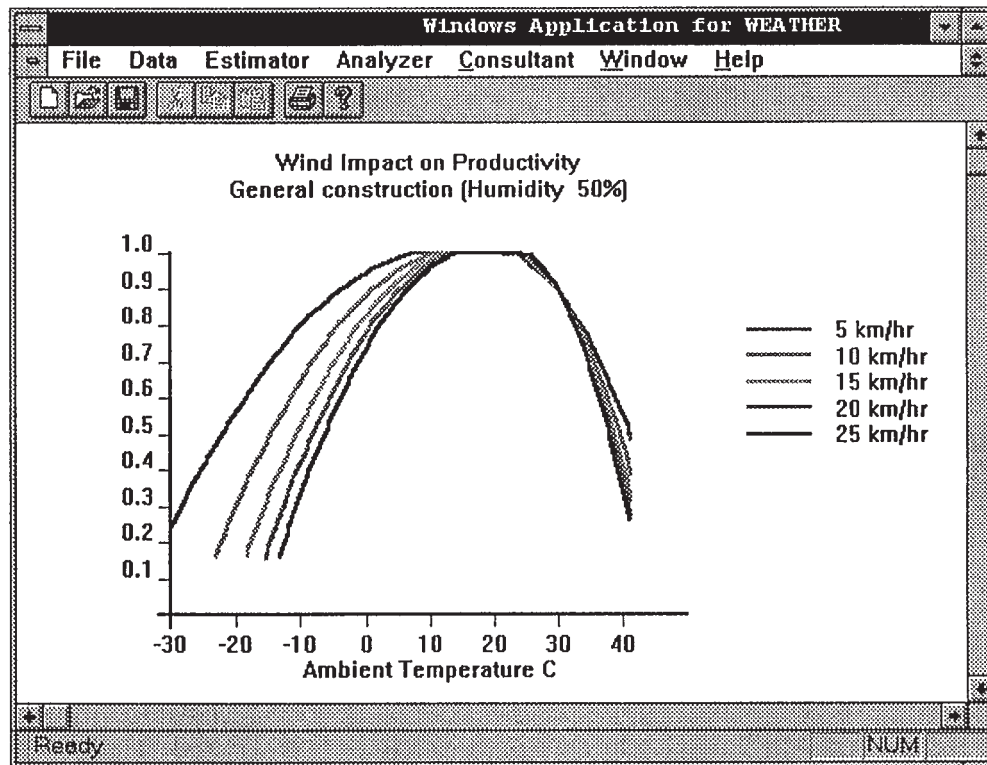
Database

The database houses two main sets of data: weather data and productivity factors. The weather set contains hourly records of temperature, humidity, wind speed, and precipitation for a 10-year period. The productivity set contains tables and formu-

las used to calculate applicable productivity factors based on temperature, humidity, and wind speed. The stored weather data depend on the location of the construction site being considered and therefore vary from one city to another. WEATHER can be used across Canada, where weather data are stored in a format consistent with that of Environment Canada. The user interface facilitates a direct transfer of Environment Canada's weather data to the system's database. The system can also be connected to a CD-ROM database for weather information.

A dynamic data interchange is designed and implemented in WEATHER for efficient data processing. The weather database for a particular city could contain approximately 450 000 data items of hourly weather data for the 10-year period considered in the system. Storing data of that size requires a large memory space. In order to overcome this problem and in an attempt to run the program efficiently, WEATHER performs a dynamic data interchange by extracting the necessary weather data from the weather database and transferring those data into appropriate system data arrays. The dynamic data interchange is carried out following the data input by the user in the two previously described dialogue windows (Figs. 3 and 4). Based on the input data of the Site Parameters dialogue window (i.e., location of the construction site and weather threshold values), WEATHER transforms the stored weather data from its original hourly values to average daily values and saves it in the system daily data arrays so as to reduce the total

Fig. 2. Productivity trends.



data items from 450 000 to approximately 25 000 items. Based on the data entered in the Activity Parameters dialogue window (i.e., specified period of scheduled construction work), WEATHER further reduces the data in the system daily arrays from a complete year to the daily data over the construction period being considered. As such, efficient management of the required memory space for running WEATHER is achieved.

Algorithm for estimating activity duration

The estimation algorithm operates automatically once a task is selected. The algorithm (i) calls the applicable formula for the selected task from the database, (ii) extracts weather data within the specified period of construction work, (iii) estimates lost working hours due to adverse weather conditions based on the specified thresholds values and the type of work interruption, and (iv) calculates, for the activity being considered, productivity factor and the mean and standard deviation of the duration. WEATHER estimates the activity duration based on the 10-year record of weather data stored in the system. For a given activity, a duration is first estimated using the weather data for each of the 10 years. This results in 10 estimated durations for that activity. These 10 estimates are then used to calculate the mean activity duration and its variance, along with the optimistic, most likely, and pessimistic activity durations.

In order to estimate an activity duration in a particular year, j , the average daily productivity factor, PF_{ij} , is calculated for each day, i , in the specified construction period. The factor PF_{ij} is calculated combining the impact of reduced labour productivity and work stoppage as follows:

$$[1] \quad PF_{ij} = PF_{ij}^{th} PF_{ij}^{in}$$

where PF_{ij} is the productivity factor considering the combined impact of reduced labour productivity and work stoppage on day i of year j ; PF_{ij}^{th} is the productivity factor considering the reduced labour productivity due to temperature, humidity, and wind speed on day i of year j ; and PF_{ij}^{in} is the productivity factor considering the effect of work stoppage due to adverse weather conditions of wind, rain and (or) snow on day i of year j .

The factor PF_{ij}^{th} is estimated based on (i) the stored weather conditions on day i of year j and (ii) available tables and formulas relating temperature, humidity, and wind speed to productivity for a number of construction tasks. For example, the tables generated by the National Electrical Contractors Association (1974), Grimm and Wagner (1974), and Koehn and Brown (1985) are used to estimate PF_{ij}^{th} for electrical works, masonry, and general construction tasks, respectively. The factor PF_{ij}^{in} is calculated as follows:

$$[2] \quad PF_{ij}^{in} = \frac{W - HL_{ij}^{wr}}{W} \times PF_{ij}^{s}$$

where HL_{ij}^{wr} is the lost working hours due to wind velocity and (or) rainfall exceeding the default or user-specified threshold values on day i of year j ; W is the total daily working hours; and PF_{ij}^{s} is the productivity factor due to snow on day i of year j .

If snow data are available on an hourly basis, lost hours due to snow can directly be integrated in the estimation of HL_{ij}^{wr} . The snow data provided by Environment Canada, however, are available on a daily basis, and therefore PF_{ij}^{s} has to be calculated independently and incorporated in the calculation of PF_{ij}^{in} . It should be noted that PF_{ij}^{s} has a binary value of 1 if the snow accumulation on that day, s_{ij} , is less than the default

Fig. 3. Site data.

Site Data

Area

Montreal

☐ Reselect data file

Weather Sensitivity Thresholds

Rain threshold:

Stop working in moderate rain

Snow threshold:

0

mm/day

Wind threshold:

25

km/h

Working Hour Losses due to weather impact

Type A:

☐ Losses equal to hours of weather conditions beyond the specified thresholds.

Type B:

☒ Losses are more than those identified in Type A.

Morning:

if

2

hours of continous rain or/and wind,

lose

☐ a half of a working day.

or

☒ a whole working day.

Afternoon:

if

4

hours of continous rain or/and wind,

lose a half of a working day.

OK

Cancel

Fig. 4. Activity data.

Activity Data

Type of construction task

General construction

OK

Cancel

Activity schedule

Ideal activity duration without considering weather influence

30

days

Activity starting date

Year

1997

Month

September

Day

1

or user-specified threshold value, s_t ; and it has a value of 0 if otherwise. That is,

[3]

$$PF_{ij}^s = 1, \text{ when } s_{ij} \leq s_t$$
$$PF_{ij}^s = 0, \text{ when } s_{ij} > s_t$$

The daily productivity factor, PF_{ij} , obtained from [1] is then used to calculate the daily effective working hours, H_{ij} , of day i in year j as follows:

[4]

$$H_{ij} = PF_{ij} W$$

where H_{ij} is the productive daily working hours on day i of year

Fig. 5. Weather data.

Weather Data

Item No. 1

Date (yyyy/mm/dd)
19970901

Recording time period

Start: 7:00

Finish: 18:00

Weather data input

Relative humidity

60

%

Wind speed

15

km/hr

Snow

0

mm/day

Temperature

25

(C)

Rain

0

raining hours

Next

Next same

Backward

Open

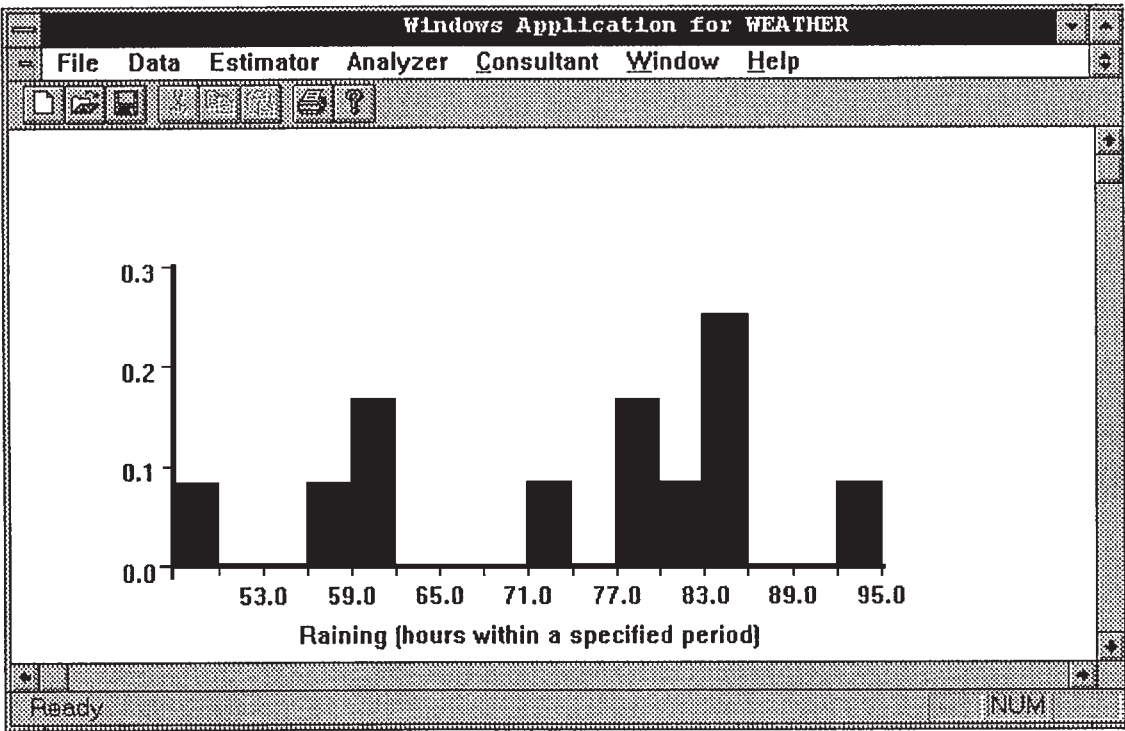
Save

Save As

Close

Cancel

Fig. 6. Statistics of rain in a specified period.

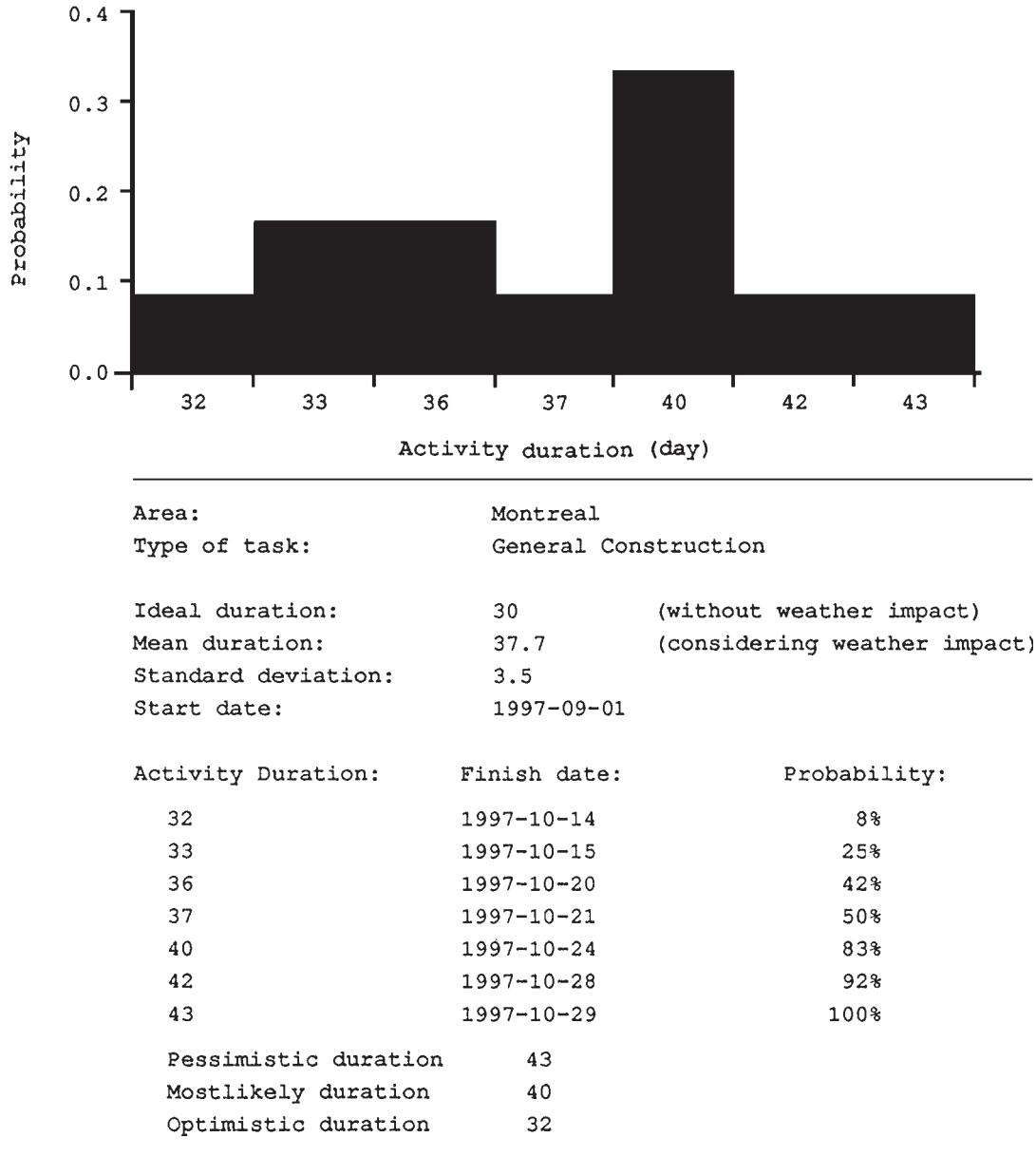


j after discounting lost hours due to weather. For example, a daily productivity factor of 0.5 ($PF_{ij} = 0.5$) affecting an 8 working-hours day ($W = 8$) results in an effective daily working hours of 4 ($H_{ij} = 4$). This means that only 4 h worth of work can be considered on day i of year j due to adverse weather conditions.

User-estimated activity duration, D , without considering weather impact, is then used to calculate the total working hours, R , required to complete the activity as follows:

[5] $R = DW$

Based on the required number of hours, R , the adjusted

Fig. 7. Analysis results.

activity duration, D_j , in year j is calculated in an iterative manner, considering only the productive daily working hours, H_{ij} , obtained from [4]. The calculations are performed as follows:

$$R = R - H_{ij}$$

[6] $i = i + 1$

$$D_j = D_j + 1$$

It should be noted that prior to the application of [6], the variables used are initialized as follows: R equals the value obtained from [5]; i is the user-specified start day of the activity (e.g., 1st of September); j is the year of the weather data being considered; and $D_j = 0$. In order to calculate D_j , the operations described in [6] are iterated until R is less than or equal zero.

For each yearly record j , an activity duration, D_j , is estimated following the above procedure. These D_j values are then used to estimate the pessimistic duration (i.e., the largest value of D_j), the optimistic duration (i.e., the smallest value of D_j), and the most likely duration (i.e., the most frequent value of D_j , or the mode in statistical terms). In addition, the mean and the variance of the activity duration are estimated as follows:

$$[7] \quad \bar{D} = \frac{1}{J} \sum_{j=1}^J D_j$$

$$[8] \quad V(D) = \frac{1}{J-1} \sum_{j=1}^J (D_j - \bar{D})^2$$

where $\bar{D}$ is the mean activity duration considering weather

impact; $V(D)$ is the variance of the estimated $\bar{D}$; and J is the number of years considered.

Numerical example

A numerical example is analyzed to illustrate the use of the system and demonstrate its capabilities. The example shows the impact of weather on an activity which involves general construction and is planned to be executed in the city of Montreal. The activity duration is estimated to be 30 days without the impact of weather and is scheduled to start on September 1, 1997 (see Fig. 4). Construction operations are to be interrupted during periods having wind velocity and (or) rainfall in excess of the threshold values of 30 km/h and moderate, respectively (see Fig. 3). In addition, a whole working day should be called off in the event of 2 or more working hours are lost in the morning due to adverse wind and (or) rainfall conditions. In this case, weather impact on the activity being considered can best be modelled by selecting type B of work interruption as shown in Fig. 3.

The impact of weather on the activity is analyzed using the system. WEATHER performs the algorithm, described earlier, for estimating the activity duration and presents an output as shown in Fig. 7. The output indicates that the estimated mean duration of the activity is 35.9 days with a standard deviation of 2.8 days. It also shows that the estimated pessimistic, most likely, and optimistic durations are 41, 36, and 31 days, respectively. In addition, the output provides the probability density of the duration in graphical and numerical forms (see Fig. 7).

Summary and concluding remarks

A decision support system, WEATHER, has been developed for estimating the impact of weather on productivity in construction. The system considers the combined impact of reduced labour productivity and interrupted construction work. WEATHER is coded using Visual C++ for Windows and provides an application that runs on Microsoft Windows. The system has three main modules: Consultant, Estimator, and Analyzer, supported by a user interface and a database. The system can be used as (i) a stand-alone module that considers only weather or (ii) a submodule that can be integrated with a scheduling system that considers other productivity-related factors such as learning curve effect and site congestion. WEATHER is portable and can be used in all Canadian cities and other cities that have a historical weather database format consistent with that of Environment Canada. The system can be applied as a useful tool in the preparation of realistic schedules before and during the construction stage, and in the analysis of weather-related claims after construction.

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List of symbols

- D user-estimated activity duration without considering weather impact
- D_j adjusted activity duration considering weather conditions of year j
- $\bar{D}$ mean activity duration considering weather impact
- H_{ij} productive daily working hours on day i of year j after discounting lost hours due to weather
- HL_{ij}^{wr} lost working hours due to wind velocity and (or) rainfall exceeding the default or user-specified threshold values on day i of year j
- J number of years considered
- PF_{ij} productivity factor considering the combined impact of reduced labour productivity and work stoppage on day i of year j
- PF_{ij}^{in} PF productivity factor considering the effect of work stoppage due to adverse weather conditions of wind, rain, and (or) snow on day i of year j
- PF_{ij}^s productivity factor due to snow on day i of year j
- PF_{ij}^{th} productivity factor considering the reduced labour productivity due to temperature, humidity, and wind speed on day i of year j
- R total working hours required to complete the activity
- $V(D)$ variance of the estimated $\bar{D}$
- W total daily working hours